

Boosting

Sample	Feature 1 (Age)	Feature 2 (Income)	Feature 3 (Credit score)	Target (Loan Approved)
S ₁	20	40	650	Yes
S ₂	30	60	700	Yes
S ₃	45	35	600	No
S ₄	50	30	580	No
S ₅	28	55	720	Yes

Target Encoding

Yes = +1

No = -1

Questions:

1. Assign initial weights to all 5 samples.
2. Construct a decision stump using feature 1 (Age) to classify the samples.
3. Identify misclassified samples and compute the weighted error rate.
4. Calculate the classifier weight.
5. Update the sample weights for the next iteration.

① Initial weights

$$w_i = \frac{1}{N} = \frac{1}{5} = 0.2$$

② Decision stump

Age < 40 → Yes (+1)
Age ≥ 40 → No (-1)

Target, (y_i)	Predict, ($h_i(x_i)$)	w_i
Yes (+1)	+1	0.2
Yes (+1)	+1	0.2
No (-1)	-1	0.2
No (-1)	-1	0.2
Yes (+1)	+1	0.2

③ Weighted error:

$$\epsilon_i = \sum w_i \cdot \mathbb{I}(y_i \neq h_i(x_i))$$

$$= [0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0 + 0.2 \times 0] = 0$$

④ Classification/Model Error:

$$\alpha_i = \frac{1}{2} \ln \left(\frac{1 - \epsilon_i}{\epsilon_i} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1-0}{0} \right) \rightarrow \infty$$

⑤ Update weight:

$$w_i^{\text{new}} \rightarrow w_i e^{-y_i \alpha_i h_i(x_i)}$$

↓

$$e^{-\infty} = 0$$

⑥ Prediction:

Model will stop training (x_i)

$$H(x) = h_i(x_i)$$

③ weighted error.

$$\begin{aligned} \epsilon_1 &= \sum w_i \cdot I(y_i \neq h_i(x_i)) \\ &= 0.2 \times 1 = 0.2 \end{aligned}$$

Target (y_i)	Predict ($h_i(x_i)$)	w_i	w_i^{New}
Yes(+1)	+1	0.2	0.16
Yes(+1)	+1	0.2	0.1
No(-1)	-1	0.2	0.1
No(-1)	-1	0.2	0.1
No(-1)	+1	0.2	0.4

④ classifier/model error.

$$\alpha_1 = \frac{1}{2} \ln\left(\frac{1 - \epsilon_1}{\epsilon_1}\right)$$

$$= \frac{1}{2} \ln\left(\frac{1 - 0.2}{0.2}\right)$$

$$= 0.694$$

w_i^{New}
0.125
0.125
0.125
0.125
0.5

⑤ Update weight:

$$w_i^{New} \rightarrow w_i e^{-y_i \alpha_1 h_i(x_i)}$$

$$0.2 \times e^{-[1 \times 0.694 \times 1]} = 0.1$$

for each sample update weight.

⑥ Normalized weight.

$$w_i^N = \frac{w_i}{\sum w_i}$$

$$w_1^N = \frac{0.2}{0.1 + 0.1 + 0.1 + 0.1 + 0.4} = \frac{0.2}{0.8} = 0.125$$

$$w_5 = \frac{0.4}{0.8} = 0.5$$

⑦ Prediction:-

$$H(x) = \text{sign} \left[\sum_{i=1}^N \alpha_i h_i(x_i) \right]$$

$$= \text{sign} [\alpha_1 h_1(x_1) + \alpha_2 h_2(x_2) + \dots]$$

→ In ~~exam~~ exam they won't go to calculate the complete Prediction.